

Yi Hu

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PERSONAL SUMMARY

PhD researcher specialized in **embodied AI** and **surgical robotics**. Developed **Vision-Language-Action (VLA) model** and **Reinforcement Learning (RL)/Imitation Learning (IL)** frameworks for Franka and da Vinci surgical robots. Strong background in SLAM, kinematics/dynamics, and control theory.

EDUCATION

PhD in Biomedical Engineer <i>University of Alberta</i>	Jan. 2021 – Exp. Apr. 2026 <i>Edmonton, Canada</i>
MSc in Control Theory and Control Engineering <i>Sichuan University</i>	Sept. 2017 – Jun. 2020 <i>Chengdu, China</i>
B.S. in Process Equipment and Control Engineering <i>Hefei University of Technology</i>	Sept. 2013 – Jun. 2017 <i>Hefei, China</i>

PROJECT EXPERIENCE

Human-Centered Autonomy Lab <i>Graduate Research Assistant, University of Alberta</i>	Jan. 2024 – Present <i>Edmonton, Canada</i>
- Built a large VLA model with 3D spatial reconstruction and mixture-of-expert (MoE) model, and validated in robotic simulator environment (Libero). VLA workflow: Collected and post-training pi0.5 model on large-scale multimodal datasets (over 300G, 5 manipulation tasks), deployed for real-time RGB-D inference on a Franka robot. - Engineered a real-time motion policy for dynamic fast manipulation by architecting a single-step diffusion model; Optimized the inference engine to achieve 7-10x lower latency; Demonstrated robustness via sim-to-real transfer (PyBullet to Franka robot) - Developed a Reinforcement Learning (RL) framework transferring human datasets priors to robotic surgical subtasks. Enabled optimized patient side manipulation and endoscopic camera operation from human daily demonstrations. Demonstrated robustness via sim-to-real transfer (PyBullet to da Vinci surgical robot)	
Telerobotic and Biorobotic Systems Group <i>Graduate Research Assistant, University of Alberta</i>	Jan. 2021 – Dec. 2023 <i>Edmonton, Canada</i>
- Built an end-to-end autonomous ultrasound system on a Franka robot. Fused classical control (admittance control) with a real-time deep learning image processing to regulate probe contact. Engineered data foundation that collects and auto-labels multimodal sensor data from force-torque sensor and video grabber. - Developed an autonomous endoscopic camera system for the da Vinci surgical robot by Imitation Learning (IL) from noisy surgical data; Engineered a safety-critical and collision-aware motion planning framework . - Developed a real-time teleoperation pipeline that transformed free-hand movements into 6-DoF robot control; Integrated a lightweight gesture classifier trigger discrete robot actions.	
Perception, Motion Control and Intelligent Robot Innovation Lab <i>Graduate Research Assistant, Sichuan University</i>	Sept. 2017 – Dec. 2020 <i>Chengdu, China</i>
- Implemented a real-time event-camera SLAM pipeline for mobile robot navigation with scene understanding, enabling 6-DoF motion tracking and 3D reconstruction through optimization-based mapping and keyframe back-end refinement. Built a physics-based modeling in Unity for mobile robot navigation. - Designed robust nonlinear control algorithms for joint-space control, accounting for actuator saturation and incorporated disturbance observers, with system stability guaranteed through Lyapunov analysis. - Derived kinematic and dynamic models for a wheel mobile robot, enabling motion planning, control design, and trajectory tracking.	

SKILLS

Programming and Frameworks: Python, MATLAB/Simulink, C++, Keras, TensorFlow, Pytorch, Linux, Ubuntu
Robotics and Tools: MuJoCo, Isaac Gym, PyBullet, ROS2, RViz, Gazebo, Git, SolidWorks, CATIA, AutoCAD, Ansys, Comsol, LaTeX
Soft Skills: Communicating cross functionally, Continuous Learning, Innovative Thinking, Event Management
Languages: Mandarin(Native), Cantonese(Native), English(Fluent)